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Ans. 1

We know the arithmetic mean,

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\Rightarrow \text{Mean} = \frac{12\times 5 + 18\times 15 + 27\times 25 + x\times 35 + 17\times 45}{80+x} \quad 6\times 55$$

$$\Rightarrow 28 = \frac{2100 + 35x}{80+x}$$

$$\Rightarrow 2240 + 28x = 2100 + 35x$$

$$\Rightarrow 7x = 140$$

$$\therefore x = 20$$

$\therefore$ The missing frequency is 20.

Ans. 2

The variance of dataset 5,5,5,5, is 0. Because, all values are identical, there is no deviation from the mean (mean = 5).

Ans. 3

$$\text{Bernoulli variance} = p(p-1)$$
$$= 0.5(1-0.5)$$
$$= 0.25$$

$$\therefore SD = \sqrt{\text{variance}}$$
$$= 0.5$$

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Ans. 4

The gradient of a scalar function $f(x,y)$ is the vector of its partial derivatives.

$$\frac{\partial f}{\partial x} = 6x + 2y \qquad \Big| \; f(x,y) = 3x^2 + 2xy + y^2$$

$$\frac{\partial f}{\partial y} = 2x + 2y$$

$$\therefore \nabla f(x,y) = (6x+2y,\ 2x+2y)$$

Ans. 5:

Given, Mean $= 12.7$

Sum $= 12.7 \times 5 = 63.5$

Let, the extra number be x.

$$\frac{63.5 + x}{6} = 13.1$$

$$\Rightarrow \quad 63.5 + x = 78.6$$

$$\therefore x = 15.1$$

Ans. 6:

We can define a variable as a float, as the following ways:

$\boxed{x = 5.0}$, or $\boxed{x = \text{float}(5)}$

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7.

```
print ("Apple", "Banana", "Cherry", sep = "-")
```

Output:

Apple, Banana, Cherry

8.

Using 'continue' statement, python can skip current iteration.

Code:

```
for i in range (5):
    if i == 2:
        continue
    print(i)
```

output:

0
1
3
4

9.

Use the 'startswith()' method in Python to check if a string starts with a specific prefix.

Code:

```
text = "Hello, world"
prefix = "Hello"
if text.startswith(prefix):
    print("The string starts with it")
```

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10.

In search algorithm, 'Completeness' refers to the property that guarantees the algorithm will find a solution if one exists or confirm no solution exists. Example:

Breadth-First-Search (BFS)

11.

Normalization	Standardization
i) It rescales data to a fixed range, usually 0 to 1.	i) Transforms data to have mean = 0, standard deviation = 1
ii) $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$	ii) $x' = \frac{x - \mu}{\sigma}$
iii) Sensitive to outliers	iii) More robust
iv) Used in image, neural networks.	iv) Used in algorithm: linear regression, SVM.

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12.

Ordinary Least Squares (OLS) is a method used in regression analysis to find the best fitting line (or model) by minimizing the sum of squared differences (errors) between predicted and actual values. OLS minimizes:

$$\Sigma (y_i - \hat{y}_i)^2$$

13:

- Feature selection means choosing a subset of the original features without changing them. Example

$$[\text{Age, Height, Weight, Salary}]$$
$$\downarrow$$
$$[\text{Age, Salary}]$$

- Feature extraction means creating new features by transforming or combining the original ones.

$$[x_1, x_2] \longrightarrow z = [x_1 + x_2]$$

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14. A chatbot is an NLP powered system that understands and responds to human language to carry out conversations automatically.

15.

Heuristic search improves upon traditional search algorithms by using a heuristic function to guide the search process toward the most promising paths, rather than exploring all possible paths blindly.

example: A* search.

16.

A single Layer Perceptron has only one layer of weights and no hidden layer, whereas multilayer perceptron consists of multiple layers of perceptrons, including one or more hidden layers.

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17.

Term frequency measures how often a word appears in a document. It contributes to feature importance by helping identify which terms are most relevant or distinctive within a specific document. / the key terms that represent the documents main topic.

18

An Artificial neural network is composed of layers of neurons (nodes). Generally three layers: Input layer, one or more hidden layer and Output layer.

19.

In a transformer, FFN is a fully connected layer applied after attention layer. It processes each position independently to add non-linearity and transformation.

20.

C# (C-sharp) is the most commonly used ~~in~~ language in Unity.

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21.

Inside-out: Sensors/cameras are on the device and track environments. e.g. Meta Quest

Outside-in: External sensors track the device. Example: HTC Vive

22.

- Navigational interfaces: Allow users to move through virtual environments (walking, flying..)
- manipulation interfaces: Allow users to interact with or modify virtual objects. (rotating, grabbing..)

23.

Cybersickness is motion sickness caused by VR/AR. It occurs due to sensory conflict - when what our eyes see (motion) doesn't match what our body feels.

24.

Mobile VR: Runs on smartphones or standalone headsets (e.g., Oculus Quest). Portable, affortable, but limited graphics and processing.

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High-end VR: Requires powerful PCs/consoles (HTC vive). Offers higher fidelity, better graphics, but expensive and less portable.

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